Technical Report on FDABench: A Benchmark for Data Agents on Analytical Queries over Heterogeneous Data

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This report provides supplementary material for FDABench, with section numbering independent of the main paper. It details the construction configuration and prompt templates, the expert verification protocol and demonstration examples, TOS validation and reproducibility settings, and extended dataset, rubric, and system descriptions.

1 System Configuration

Configuration Parameters

Construction Agent Model	
Model	anthropic/claude-opus-4.5
Temperature	0.7
Max Output Tokens	1,000 (planning), 1,800 (report)
Web Search Model	
Model	perplexity/sonar
Max Output Tokens	4,000 tokens
Temperature	0.7
Embedding Model	
Model	text-embedding-3-small
Dimensions	1,536
Batch Size	50
Vector Index	
Type	FAISS Index
Metric	Cosine Similarity
Index Size	86,000+ domain knowledge chunks
Top-k Retrieval	5
Chunk Size	512 tokens
Evaluation Model	
Model	google/gemini-3-flash-preview
Max Output Tokens	500
Validation	
Expert Pool	6 experts

2 Prompt Templates

Web Search Prompt (Abstracted)

Input: {query}

Goal: Retrieve authoritative external evidence that complements structured database facts (e.g., standards, regulations, market/clinical reports) and provides citations usable for validation and reporting.

Expected output:
- A concise, structured summary of findings
- Key statistics and data points (when available)
- Recent trends or developments relevant to the query
- Traceable citations (URLs or publication identifiers)

Source preferences: Peer-reviewed journals; industry reports from recognized consulting firms; government publications; and verified business sources with traceable citations.

Chain Planning Prompt (Abstracted)

Inputs: {query}, {sql_result}

Available tools: `web_search` (external validation and current context), `vector_search` (domain knowledge and methodology), `file_search` (local file system for domain-specific documents), `db_explore` (database schema, table relationships, and sample data).

Planner objective: Propose a short sequence of external retrieval steps that (i) is conditioned on the SQL result and previously retrieved evidence, and (ii) avoids redundant queries while covering missing context needed for synthesis.

Output format: A JSON array of steps with fields {`step`, `tool`, `query`, `rationale`}. For hard tasks, the planner typically produces 2–4 steps.

Vector Database Retrieval

Retrieval Strategy:
(1) Generate vector embedding using OpenAI text-embedding-3-small
(2) Query FAISS index with 86K+ domain knowledge chunks
(3) Retrieve top-5 most semantically similar document chunks
(4) Return chunk metadata (category, file name, score) and text preview

Document Priorities: Foundational theoretical frameworks; technical specifications and standards; case studies and empirical analyses; domain-specific knowledge bases; regulatory and institutional documents.

Report Generation Prompt (Abstracted)

Inputs: {enhanced_query}, {sql_result}, {web_context}, {vector_context}

Goal: Produce an analytical report that grounds claims in database evidence and integrates complementary external knowledge. To support rubric-based evaluation, the prompt specifies a fixed section schema.

Report structure (section schema):
- **Executive Summary:** Summarize the main finding from SQL data and why it matters.
- **Data Analysis Results:** Interpret SQL results with concrete numbers and statistics.
- **External Context & Insights:** Incorporate external evidence (web/vector) that contextualizes or validates the SQL findings.
- **Key Connections:** Explain how database patterns connect to external mechanisms, policies, or domain frameworks.
- **Conclusions:** Provide final conclusions and (when applicable) actionable implications.

Grounding guidance: Prefer claims that can be traced to SQL outputs or cited external evidence; avoid unsupported assertions.

Enhanced Query Generation Prompt (Abstracted)

Input: {original_query}

Objective: Rewrite the original query into a more realistic analytical request that encourages multi-step reasoning and multi-source synthesis, while avoiding leakage of intermediate results (e.g., explicit numbers from demonstration SQL execution).

Output: A 2–3 sentence enhanced query describing the analysis goal and required synthesis at a high level.

Choice Task Construction (SC/MC)

Grounding principles:
- The correct option(s) are verifiable from executable SQL evidence plus complementary external context.
- Distractors reflect realistic analytical errors (e.g., misinterpretation, incomplete integration) rather than arbitrary trivia.
- Questions emphasize data interpretation and synthesis instead of requiring specialized background knowledge not provided in $\mathcal{I}$.

Answer Option Design:
- Correct option(s): directly supported by SQL + external evidence.
- Distractors: plausible but incorrect due to (i) misreading the SQL result, (ii) ignoring external context, or (iii) overclaiming beyond evidence.

3 Expert Verification Protocol

Expert Verification Guidelines

Expert Pool: 6 PhD-level experts in database systems and analytics, each with 5+ years of SQL proficiency across multiple dialects and familiarity with enterprise analytical scenarios.

Verification Criteria:

1. Integration Necessity (1-5 Likert scale, threshold $\geq$ 4):
- Whether multi-source integration provides semantic enrichment
- Presence of causal relationships across data sources
- Contextual validation beyond single-source facts

2. Answer Correctness:
- SQL re-execution for numerical verification
- Source cross-referencing for factual accuracy
- Reasoning chain validation for logical consistency
- Any discrepancy triggers REVISE with specific feedback

3. Realism Assessment (1-5 Likert scale, threshold $\geq$ 4):
- Genuine business need or research scenario
- Industry relevance and current applicability
- Appropriate complexity for target difficulty level

Decision Protocol:
- **ACCEPT:** All criteria met
- **REVISE:** Improvements needed (provide specific feedback C_k)
- **REJECT:** Fundamental flaws (single-source solvable, unrealistic)

Disagreement Resolution: Majority voting ($\geq$ 2/3 agreement); senior expert arbitration for conflicts; inter-annotator agreement validated through Krippendorff's $\alpha > 0.78$.

4 Demonstration Data Agent Examples

The following examples demonstrate data agent scenarios that guide our dataset construction process.

Expense Reimbursement Compliance Review

An Expense ERP Data Agent conducts intelligent compliance reviews through heterogeneous data integration:

Structured Database Query:
- Retrieve employee expense claim information (employee ID, grade level, expense amount, timestamp, category, approval status)
- Execute aggregations to identify statistical patterns and outliers

Vector Database Retrieval:
- Query applicable reimbursement policies based on employee grade and expense category
- Extract allowable expense limits, required documentation, and approval workflows

External Web Validation:
- Cross-validate supporting documentation with regulatory databases
- Verify merchant information and transaction authenticity

Heterogeneous Integration:
- **SQL provides:** expense amounts, submission timing, approval rates
- **Vector database provides:** allowable limits, compliance rules
- **Web sources provide:** regulatory updates, merchant verification

Anomaly Detection: Late submissions, over-claiming beyond limits, missing documentation, policy violations based on regulatory updates.

This demonstrates genuine heterogeneous integration where **no single source is sufficient**.

Employee Contract Verification

A Contract Verification Data Agent conducts intelligent inspections of labor agreements:

Structured Database Query:
- Retrieve employee information (ID, grade level, department, employment date, contract history)
- Execute joins to correlate employee grades with regulatory requirement categories

Vector Database Retrieval:
- Query grade-specific regulatory knowledge bases
- Retrieve compliance requirements: minimum wage thresholds, probation lengths, leave policies

External Web Validation:
- Cross-validate contract clauses with recent labor law amendments
- Verify minimum wage standards against regional labor authority announcements

Heterogeneous Integration:
- **HR database provides:** employee facts, grade levels, department assignments
- **Contract documents provide:** compensation clauses, probation durations
- **Vector database provides:** grade-specific standards, required salary ranges
- **Web sources provide:** current legal context, regulatory updates

Compliance Issues: Below-standard compensation, excessive probation periods, missing legal disclosures, outdated contract templates.

This demonstrates multi-modal heterogeneous integration where **each data source provides essential non-redundant information**.

5 Task DAG and TOS Validation

To verify that TOS generalizes beyond construction-time tool preferences, we randomly sample 150 agent execution traces (50 per difficulty level) produced by five target systems. Three experts independently score each trace for reasoning correctness and path validity on a 0 to 1 scale. We then compute agreement between expert consensus scores and automated TOS scores.

Table 1: TOS vs. human expert agreement on agent trace evaluation (150 sampled traces).

Comparison	Krippendorff's α	ICC(A,1)	τ_b
Human vs. Human	0.77	0.80	0.90
Human vs. TOS	0.72	0.75	0.87

As shown in Table 1, TOS achieves $\tau_b = 0.87$ against human consensus, confirming that DAG-based scoring preserves system rankings. Experts identified 23 traces (15.3%) containing valid alternative paths not covered by existing `ALT_GROUP` annotations; incorporating these improved agreement to $\alpha = 0.75$. Tool-name alias resolution employs a curated synonym table of 34 alias pairs with embedding-based fallback (cosine similarity ≥ 0.85) for unseen tool names.

6 Evaluation Reproducibility

FDABench ensures reproducibility through: (1) frozen retrieval snapshots per task, eliminating live API dependence; (2) fixed judge configuration with temperature 0 and versioned model; (3) deterministic primary checks via exact match for SQL and choice tasks; and (4) versioned evaluation code. We verify stability by running the full pipeline three times on 200 tasks with identical configuration: choice-based metrics (EX_SC, EX_MC) are perfectly deterministic as they rely on exact string matching; LLM-judged metrics (RS, TOS) exhibit CV $< 2\%$ across runs, confirming that temperature-0 inference with constrained output format produces stable scores.

7 Data Collection Criteria

For each candidate query from the original datasets containing over 23,900 queries, we manually curated relevant unstructured files by searching authoritative sources (e.g., Google Scholar, arXiv) based on the database topic and query intent, yielding over 1,600 files. Files are retained if they satisfy:

- **Complementarity:** Providing causal explanations or contextual validation not derivable from database facts alone (e.g., financial regulations for transaction analysis, medical guidelines for patient outcome assessment, market reports explaining sales trends).
- **Authority:** Publicly accessible peer-reviewed publications or institutional documents only (arXiv preprints, government publications, open-licensed educational content, institutional white papers; exclude user-generated content without editorial oversight). Licensing details are provided in Section 15.
- **Coverage:** Balanced representation across finance, healthcare, retail, and manufacturing sectors with multiple perspectives per domain.

The collected corpus includes five categories: academic literature (theoretical frameworks), industry documentation (market analyses, regulatory documents), multimedia content (educational videos, expert seminars, and domain-specific figures), technical specifications (API documentation), and enterprise knowledge (real application case studies).

Processing Pipeline. Collected data undergoes: (1) format normalization, (2) chunking (512-token chunks with sentence-boundary-aware splitting), (3) embedding generation (text-embedding-3-small, 1,536 dimensions), (4) index construction (FAISS index with cosine similarity), and (5) quality verification.

8 Data Modality and Processing

Table 2 summarizes the unstructured modality composition across tasks. A single task may involve multiple modalities, so percentages overlap.

Retrieval operates through two complementary channels. *Vector retrieval* indexes textual content via FAISS for semantic search, while *file system search* provides access to raw multimodal assets (images, audio, video) through metadata. By default, all multimodal source files are provided in their original format so that systems with native multimodal capabilities can ingest them directly. Only for target systems that do not natively support a given modality (e.g., text-only LLM backends), we additionally supply pre-computed text-normalized representations such as Whisper-large-v3 transcripts

Table 2: Unstructured data modality breakdown. Task percentages overlap across modalities.

Source Modality	Files	Chunks	Tasks Involving (%)
PDF (papers, reports)	1,412	73,500+	77.8%
Image (figures, charts)	246	—	25.6%
Audio	108	7,400+	21.0%
Video	80	5,100+	14.3%
Web search	—	—	64.3%

for audio/video and OCR extractions for scanned documents, ensuring evaluation fairness without constraining multimodal-capable agents.

9 Evaluation Rubric Design

Each task instance includes a structured evaluation rubric with weighted scoring dimensions and verification methods.

Evaluation Dimensions (Weight=0.25 each)

- **SQL_ACCURACY**: Executes SQL and interprets result (Exact match)
- **EXTERNAL_INTEG**: Integrates web/vector with SQL (LLM judge)
- **LOGICAL_REASON**: SQL → context → insight (LLM judge)
- **COMPLETENESS**: Addresses all query aspects (Report check)

For hard tasks, the rubric additionally includes a **chain validation** component that verifies whether the reasoning followed the expected multi-step dependency chain. Each step records its tool type and rationale, enabling fine-grained analysis of where agents deviate from optimal reasoning paths.

10 Difficulty Classification

Following BIRD [2], task difficulty is scored by an LLM on four dimensions rated 1–5: SQL complexity, source diversity, reasoning depth, and domain knowledge. SQL complexity ranges from a simple `SELECT` (1) to multi-join subqueries with aggregation and window functions (5); source diversity ranges from SQL only (1) to SQL combined with web, vector, file, and database exploration (5); reasoning depth ranges from a direct lookup (1) to a multi-step chain with cross-source inference (5). Their average yields Easy (≤ 2.0), Medium (2.1–3.5), and Hard (≥ 3.6). When LLM scoring fails, a heuristic fallback based on the number of distinct successful source types is applied, and the final label combines expert votes weighted by the reasoning-token distribution from the LLM analysis.

11 Quality Control Statistics

Iterative Refinement Statistics

First-Iteration Acceptance	
Accepted Without Revision	1,523 tasks (75.9%)
Second-Iteration Acceptance	
Accepted After One Revision	367 tasks (18.3%)
Third+ Iteration Acceptance	
Accepted After 2+ Revisions	117 tasks (5.8%)
Efficiency Metrics	
Average Iterations per Task	1.3 iterations
Average Review Time per Task	8.3 minutes

Table 3: Rejection reasons distribution during dataset construction.

Rejection Reason	Count	Percentage
Single-Source Solvability	1,058	49.9%
SQL Errors	558	26.3%
Insufficient Integration	322	15.2%
Unrealistic Scenarios	182	8.6%
Total Rejected	**2,120**	**100%**

Final Distribution: Task types: Report 33.28% (668), Single-Choice 28.85% (579), Multiple-Choice 37.87% (760). Difficulty: Easy 20.68% (415), Medium 32.84% (659), Hard 46.49% (933).

12 Data Agent Architecture

Figure 1 illustrates the architecture of data agent systems targeted by FDABench. The architecture comprises three layers: (1) **Self-Decision Agents** implementing four workflow patterns (Planning, Reflection, Tool-use, Multi-agent); (2) **Agent Tools** including file reader, web search, vector search, semantic operators, and text-to-SQL components; (3) **Heterogeneous Data Sources** spanning relational databases, file servers, and web servers, supported by infrastructure including VectorDB, LLM, and memory systems.

13 Data Agent Workflow Patterns

FDABench encompasses four representative workflow patterns [3, 5] as illustrated in Figure 2.

Planning agents decompose complex queries into structured action sequences, reducing computational complexity through systematic task breakdown and parallel execution. **Tool-use agents** directly integrate external systems from $\mathcal{T}$, minimizing latency through immediate tool invocation. **Reflection agents** employ iterative self-evaluation where evaluative components refine outputs through multiple reasoning cycles, trading computational overhead for improved accuracy. **Multi-agent systems** distribute specialized roles across collaborative agents, handling complex heterogeneous tasks through structured communication protocols. These patterns enable comprehensive evaluation of architectural trade-offs in processing heterogeneous analytical workloads.

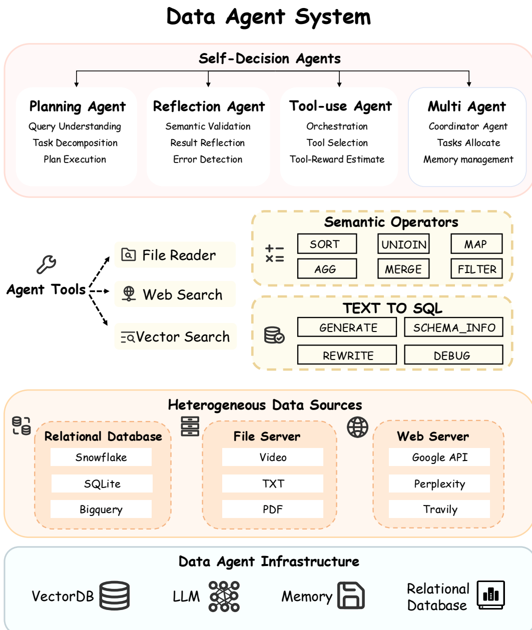

Figure 1: Architecture Overview of Data Agent System.

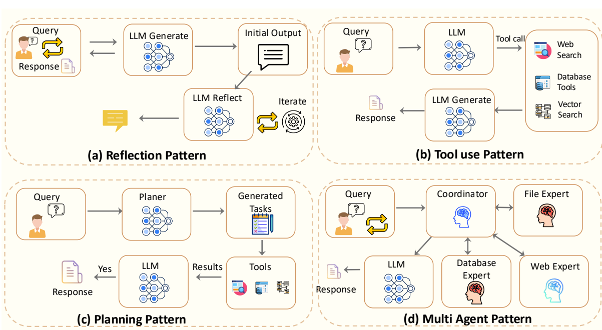

Figure 2: Four Data Agent Workflow Patterns

14 Target System Descriptions

Detailed descriptions of the evaluated target systems, spanning general analytical query systems, semantic operator query systems, and RAG systems, are provided in the appendix of the main paper.

15 Data Licensing and Ethical Considerations

Structured data. FDABench inherits structured databases from Spider [4] (CC BY-SA 4.0), BIRD [2] (CC BY-SA 4.0), Spider 2.0 [1] (CC BY-SA 4.0), and DABStep (MIT). All permit academic redistribution.

Unstructured data. The corpus is curated from publicly accessible sources: arXiv preprints (arXiv non-exclusive license), government/institutional publications (public domain), and open-licensed educational content (CC BY). We do not redistribute copyrighted full-text from commercial publishers; where such content is referenced, we provide retrieval scripts pointing to original sources and include only brief excerpts consistent with fair-use principles.

Release artifacts. We release: (1) all database schemas and instances, (2) the FAISS vector index with pre-computed embeddings, (3) frozen retrieval snapshots per task enabling deterministic evaluation without network access, and (4) complete evaluation code and prompts.

Ethics. No personally identifiable information is present. Demonstration scenarios (Section 4) are illustrative and do not reference real individuals. All annotators are co-authors or compensated research assistants.

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